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DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORKS

AIM :

To calculate and verify Z parameters of a two port network using Multisim simulator.

APPARATUS REQUIRED :

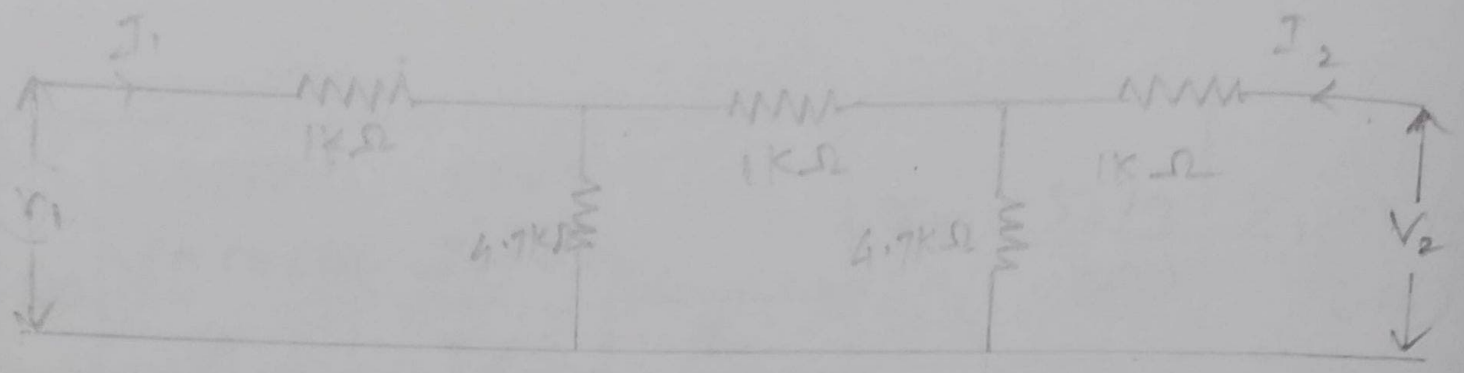
S.No	COMPONENTS	SPECIFICATION	QUANTITY
1.	Power supply		1
2.	Multimeter	-	1
3.	Resistors	10kΩ, 10kΩ, 4.7kΩ	
4.	Breadboard	-	1
5.	Connecting wires	-	-
6.	PC with Multisim simulator.	-	-

PROCEDURE :

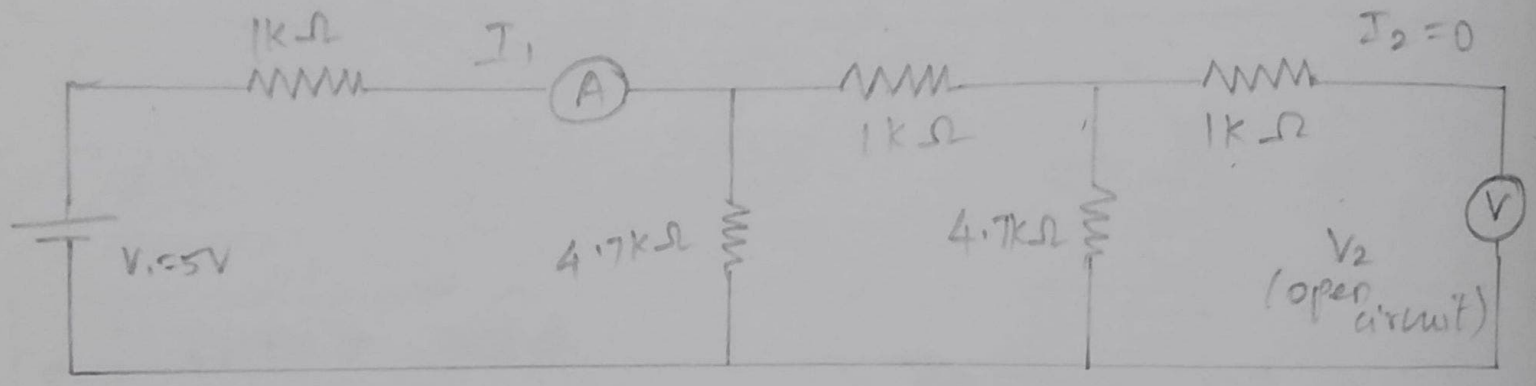
Connect the circuit as shown in fig and switch 'ON' the experimental board.

First open O/P terminal & supply 5V to I/P terminal. Measure O/P voltage and I/P current.

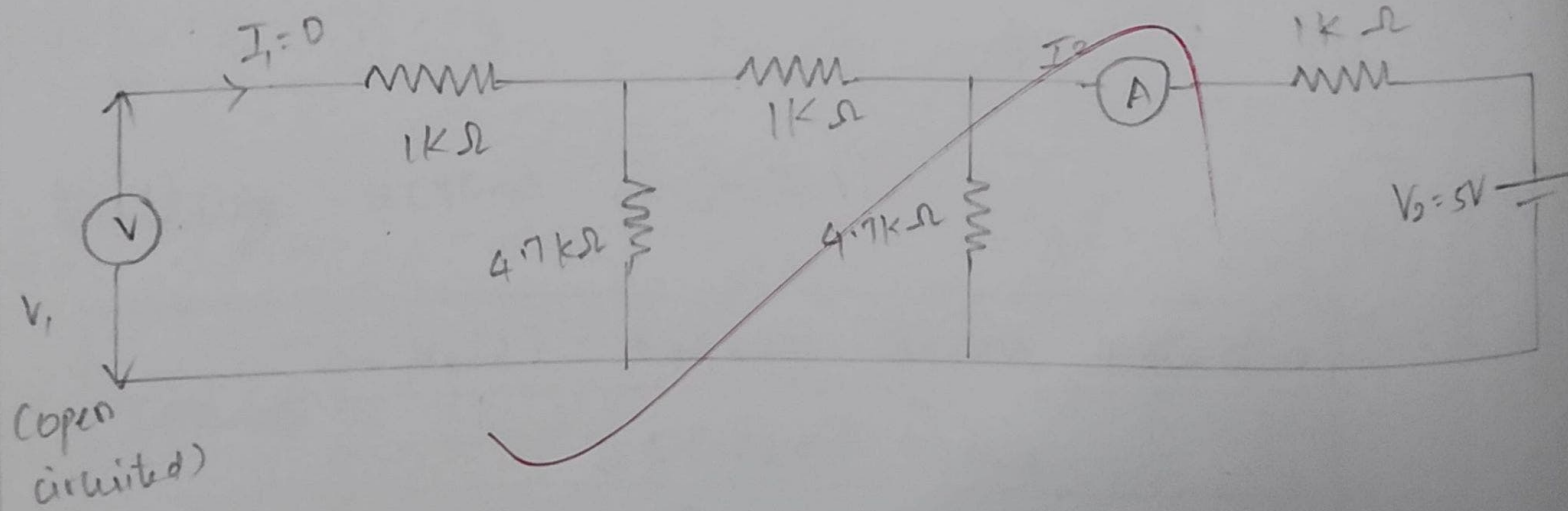
CIRCUIT DIAGRAM:



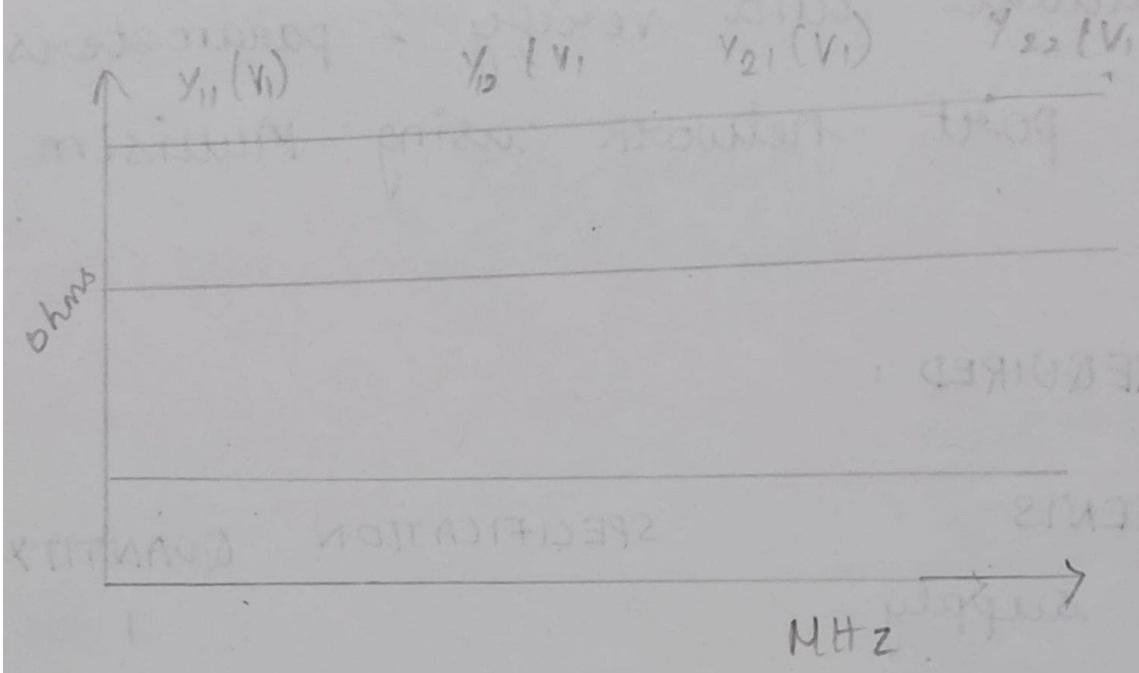
OUTPUT OPEN CIRCUITED:



INPUT OPEN CIRCUITED:



GRAPH:



OBSERVATION:

S. NO	OUTPUT OPEN CIRCUITED ($I_2=0$)			$Z_{21}=V_2/I_1$ (Ω)	$Z_{11}=V_1/I_1$ (Ω)
	V_1	V_2	I_1		
1.	1.	1	1	4	6

S. NO	INPUT OPEN CIRCUITED ($I_1=0$)			$Z_{12}=V_1/I_2$ (Ω)	$Z_{22}=V_2/I_2$ (Ω)
	V_2	V_1	I_2		
1.	1	1	1	4	10

Secondly open I/P terminal & supply 5V to O/P terminal. Measure I/P voltage & O/P current using multimeter.

Calculate the values of Z parameters using Equation (1) & (2).

Switch 'OFF' the supply after taking the readings.

MARKS SPLIT UP:

S. NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	Experimental setup / connections	20	20
2.	Observation / Readings	20	20
3.	Calculation / Result	20	20
4.	Viva	20	20
5.	Record	20	20
		Total	100

RESULT:

The Z-parameter of the two port network has been calculated and verified by using Multisim Simulator.